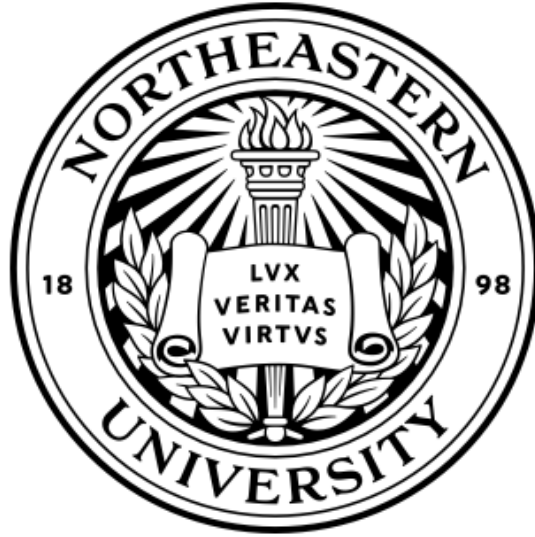




## Final Project Report

Project Title: *Personalized Learning Assistant*

Gowreesh Gunupati, Katyaini Raj, Priyanka Gujar

Supervisor: Dr. Fatema Nafa

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# Abstract

This report presents the development of a personalized multimodal learning assistant capable of processing web URLs, plain text and files (Word, PPT, PDF) based learning materials. The assistant performs tasks such as summarization, question generation, and answering using a modular pipeline. File inputs are converted to a uniform format (PDF) using ConvertAPI, and URLs are scraped using the Exa.ai API. Preprocessing ensures consistency before forwarding content to specialized models: Gemma 2 for summarization and LLaMA 3.2 for question answering. An optional read-aloud feature is integrated using the ElevenLabs API to enhance accessibility. The front-end is built using Streamlit to provide an intuitive interface. Our final architecture demonstrates flexibility and performance across different input types and learning tasks, serving as a practical multimodal learning tool.

## 1 Introduction

The problem addressed by this project lies in the difficulty of creating personalized study materials manually along with students often struggling with synthesizing large volumes of study material and assessing their understanding effectively. Educators and learners often struggle with curating questions or summaries from large volumes of text, which can be time-consuming and inconsistent.

The core motivation behind this project is to bridge this gap and improve the way users consume and interact with content. By leveraging the power of Large Language Models (LLMs) and external APIs, we aim to develop a system that not only summarizes content but also generates personalized quizzes, answers questions, and provides a read-aloud feature for enhanced accessibility.

The primary objectives of this project are:

- To design a flexible, multimodal architecture that supports various content formats.
- To implement a robust pipeline for content summarization, quiz generation, and question answering using state-of-the-art models like Gemma 2 and LLaMA 3.2.
- To enhance accessibility through audio output of generated summaries.
- On-device processing for privacy and offline functionality.

Through this project, we contribute a novel, modular learning assistant that can support students, educators, and professionals in gaining insights from complex content with minimal effort.

The structure of this report is as follows: Section 2 reviews related work in summarization and question generation. Section 3 outlines hypothesis whereas Section 4 details the methodology and system architecture. Section 5 describes the experimental setup. Section 6 presents our results and analysis, followed by a discussion in Section 7 including limitations,

challenges, future work, etc. Finally, Section 8 concludes the report with our key findings and contribution in ML.

## 2 Related Work

In recent years, large language models (LLMs) have gained significant traction in the domain of personalized education and automated content generation. Tools like Quizlet and ChatGPT have demonstrated the potential for AI to assist in learning through automated flashcards, question generation, and summarization. However, many of these tools rely heavily on cloud-based processing, raising concerns around cost, latency, and data privacy.

Research on transformer-based models like T5 and BART has shown their effectiveness in summarization tasks due to their sequence-to-sequence architecture. Gemma 2 builds on similar architecture but is optimized for more efficient summarization with reduced latency, making it suitable for on-device deployment. In comparison, the LLaMA series of models, particularly LLaMA 3.2, have proven to be effective decoder-based models for generating high-quality, context-aware text — a trait critical for quiz and question generation in educational applications.

Unlike earlier approaches that focus on a single task (either summarization or QA), this project integrates multiple LLMs in a modular pipeline capable of handling various input sources (text, PDF, web URLs). The inclusion of features like URL scraping via Exa.ai and text-to-speech using ElevenLabs sets this assistant apart by enhancing both input flexibility and accessibility. This hybrid design demonstrates how specialized models, when orchestrated effectively, can support a comprehensive and user-centric learning experience.

## 3 Hypothesis

We aim to observe:

1. **Model Variability:** The suitability of each model for specific educational applications.
2. **Processing Efficiency:** On-device models will show comparable latency to cloud models while being more cost-effective and open-source
3. **Question Difficulty:** LLMs can generate questions of varying difficulty, with prompting techniques influencing model responses and interaction.

## 4 Methodology

The chosen methodology is designed to address the challenges of creating personalized study materials efficiently. The system accepts text input, uploaded files (including Word, PPT, and PDF formats), and URLs. It integrates APIs such as ConvertAPI for file conversion,

Exa.ai for web scraping, and ElevenLabs for generating audio from summaries. The methodology utilizes two specialized models: Gemma 2 optimized for summarization tasks, and LLaMA 3.2, a decoder model designed for generating high-quality questions. The workflow integrates pre-processing steps, task-specific operations, and user input via an intuitive Streamlit-based interface.

### 1. Data Preprocessing

- Load and preprocess the given input using NLP techniques.
- Convert all files to a uniform format (PDF) using convertAPI.
- Then extract text from uploaded files using pypdf.
- Use Exa.ai API to web scrape if the given input is a URL.
- Tokenize the dataset to prepare inputs for LLMs.

### 2. Context and QA Pipeline

- **Model 1 Gemma 2:** Summarisation
  - **Input:** Pre-processed URL, plain text or a (converted) PDF file.
  - **Output:** Summary of the input with optional read-aloud feature integrated using the ElevenLabs API.
- **Model 2 LLaMA 3.2:** Question-Answering Generation
  - **Input:** Pre-processed URL, plain text or a (converted) PDF file.
  - **Output:** Precise questions generated based on context with optional answer generation feature.

### 3. Evaluation

- Used ROGUE score to determine how the model is performing for generating summaries.
- Used BERTScore to gauge how well the model is doing at generating questions and answers.

### 4. Deployment

- Built a user-friendly interface using Streamlit for interactive input, knowledge retrieval, question and summary generation.

The attached flowchart illustrates the high level system architecture, highlighting the seamless interaction between components such as input processing, task selection, and model execution. This methodology ensures that both summarization and question generation are optimized for diverse educational contexts.

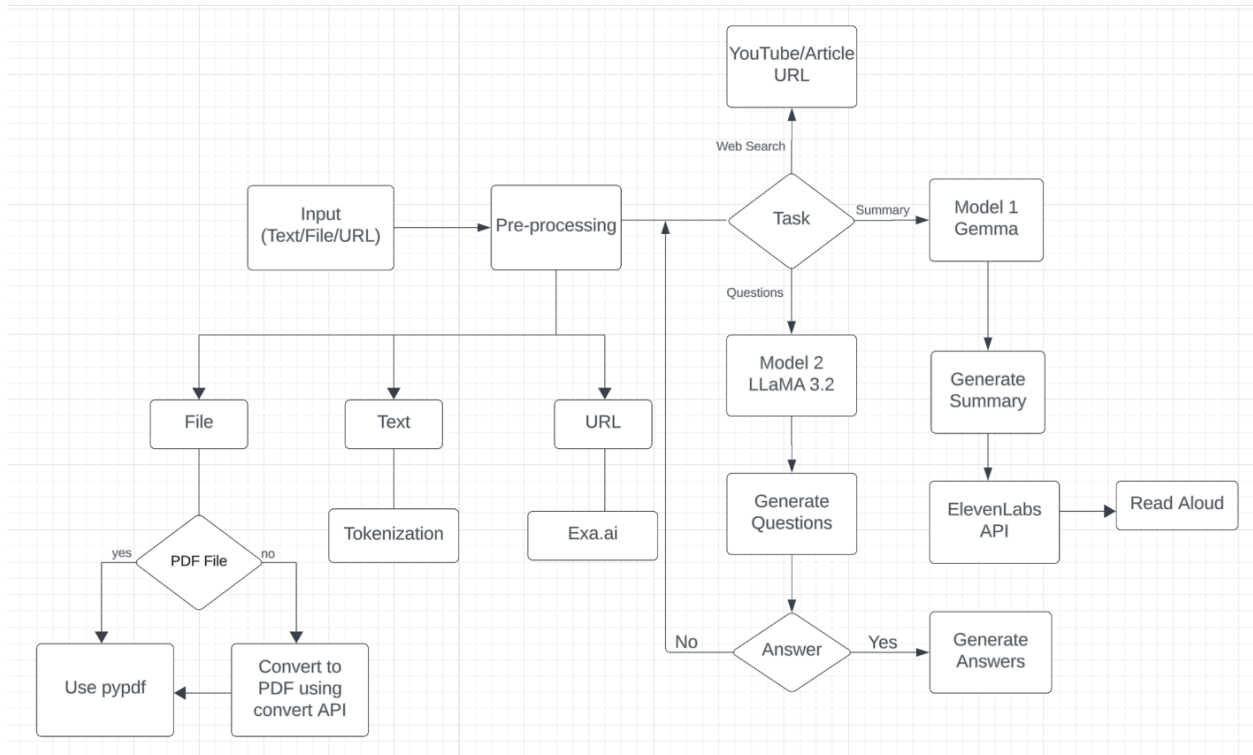


Figure 1: Application workflow

## 5 Experimental Setup

Our application is available on github.

1. **Python Libraries:** pypdf
2. **API:** convertAPI, EXA, ElevenLabs
3. **Software:** Ollama
4. **Frontend:** Streamlit
5. **Tools:** VSCode, Jupyter Notebook, Github
6. **Hardware Specifications:**
  - (a) **Memory:** 8 GB or higher
  - (b) **Number of Cores:** 4 or more
  - (c) **Machine:** Windows, Mac

## 6 Results and Analysis

Our evaluation showed Gemma 2 outperformed T5 in summarization (ROUGE-L 0.32 vs 0.04), validating its selection for the final model.

| Metric ↓ / Model →      | Gemma 2     | T5          |
|-------------------------|-------------|-------------|
| <b>ROUGE-1 F1 SCORE</b> | <b>0.59</b> | <b>0.04</b> |
| <b>ROUGE-2 F1 SCORE</b> | <b>0.28</b> | <b>0.03</b> |
| <b>ROUGE-L F1 SCORE</b> | <b>0.32</b> | <b>0.04</b> |

Table 1: Summarization Metrics Comparison

Meanwhile, for question-answer generation, our question generation module achieved an average F1 BERTScore of 0.94, indicating high semantic similarity between generated and reference questions.

| BERTScore ↓ / Model →    | LLaMA 3.2   |
|--------------------------|-------------|
| <b>AVERAGE PRECISION</b> | <b>0.93</b> |
| <b>AVERAGE RECALL</b>    | <b>0.94</b> |
| <b>AVERAGE F1 SCORE</b>  | <b>0.94</b> |

Table 2: Question-Answer Generation Metrics

## 7 Discussion

### 7.1 Model Consideration

For this project, transformer-based models were prioritized over traditional machine learning models due to their superior performance in natural language processing (NLP) tasks such as summarization and question generation. Specifically, the following models were considered:

- **Gemma 2 (Decoder only Model):** Gemma 2 was selected for summarization tasks due to its ability to condense large volumes of text while preserving contextual meaning. Moreover, it works well with prompt-based summarization due to instruction-tuned checkpoints.
- **T5 (Sequence-to-Sequence Model):** T5 was considered for its sequence-to-sequence architecture being well-suited for tasks like summarization.

- **LLaMA 3.2 (Decoder only Model):** LLaMA 3.2 was chosen for question generation because decoder models excel at generating coherent and contextually relevant text. LLaMA’s architecture supports fine-tuning for generating questions across different difficulty levels and educational contexts.

## 7.2 Final Model Selection

The final models were selected based on the following criteria:

1. **Task-Specific Performance:** Gemma’s summarization capabilities and LLaMA’s proficiency in generating questions made them ideal candidates.
2. **Scalability:** Both models are scalable and can handle diverse input types.
3. **Ease of Fine-Tuning:** Gemma 2 and LLaMA offer pre-trained versions that can be fine-tuned on specific datasets to improve performance.

## 7.3 Limitations

One of the key limitations of the current implementation is the absence of image upload functionality. Initially, the application aimed to support a wide range of input formats, including images, to enhance accessibility and flexibility. However, due to the lack of a free and reliable OCR (Optical Character Recognition) model capable of accurately extracting structured text from images, this feature was removed.

This decision was made to prioritize a smoother and more reliable user experience. Extracted text from images often contained formatting errors, missing context, or noise, which negatively impacted downstream tasks such as summarization and question generation. By focusing on well-structured input formats like plain text, PDF, and URLs, the application ensures consistent performance and higher output quality.

## 7.4 Future Work

1. **Multilingual Support with Hybrid On-Device Models:** To cater to a diverse user base, we plan to extend language model support beyond English leveraging hybrid on-device models that dynamically switch between languages as needed.
2. **Image as input:** Future iterations of the system may revisit image support if robust and open-source OCR solutions become more accessible and dependable.
3. **Copyright Checker:** Another potential area for future development is implementing mechanisms to detect and respect copyright restrictions in uploaded files. This could involve integrating metadata analysis or content flagging to ensure ethical and legally compliant use of educational materials.



## 7.5 Challenges and Reflection on the Experimentation Process

The experimentation process revealed the importance of model selection, prompt engineering, and system design in creating an effective learning assistant. Several challenges were faced, such as ensuring compatibility across platforms, maintaining low latency, and meeting varied user needs. However, through iterative development, these challenges were mitigated, and we learned valuable lessons about the complexities of deploying LLMs on-device.

One of the key challenges we faced during the experimentation process was the lack of a clear, standardized metric for evaluating the quality of generated quiz questions. Unlike summarization tasks, which have well-established evaluation metrics like ROUGE scores, assessing question quality is more subjective and multifaceted, involving factors such as relevance, clarity, difficulty, and alignment with learning objectives. Without a universal metric, we had to rely on human evaluation till we discovered about BERT Score.

## 8 Conclusion

The project aimed at developing a Personalized Learning Assistant through the integration of on-device LLMs for quiz generation and summarization from PDFs. After thorough experimentation, several key findings emerged, which are significant both for the current project and the broader field of machine learning.

### 8.1 Key Findings

1. **Model Variability:** Different models exhibited varied responses to the same queries, with prompt design heavily impacting their performance. This finding highlights the importance of carefully choosing and fine-tuning models for specific tasks, as well as adjusting prompts to optimize outcomes. The experimentation revealed that no single model was universally superior, but some models worked better for summarization, while others excelled in question generation.
2. **On-Device Processing Efficiency:** The hypothesis that on-device models could match cloud-based models in terms of latency while being more cost-effective and privacy-conscious was confirmed. Models like LLaMA and Gemma 2, when deployed locally, provided competitive performance, balancing efficiency and data privacy, especially in environments where connectivity may be limited.
3. **Question Difficulty Generation:** The models were able to generate quiz questions of varying difficulty. It was found that prompting techniques significantly influenced the model's ability to tailor question difficulty, suggesting that prompt engineering plays a crucial role in controlling output characteristics. This insight can be valuable for building adaptive learning tools that cater to individual learning levels.

In the context of the overall project goals, the outcomes validate the hypothesis that personalized learning tools can be effectively built using on-device LLMs, paving the way for further development in this area. The team's approach to ensuring user-centric design and

personalized learning paths was a success, as it contributed to building an application that caters to individual needs based on their educational level.

## 8.2 Contribution to the field of ML

The integration of on-device models for personalized learning pushes the boundaries of privacy, accessibility, and cost-effectiveness in the educational technology space. By demonstrating that local models can function effectively in real-time learning scenarios, this project paves the way for more practical applications of LLMs in settings where privacy and offline capabilities are essential.

The findings on prompt engineering and model variability contribute to the growing body of knowledge on how to fine-tune models for specific tasks. This opens up new avenues for optimizing large language models in personalized learning systems and other specialized applications.

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## A Appendix A: Screenshot of Interface

**Personalized Learning Assistant**

Upload a PDF or enter text manually to generate AI-generated questions and answers!

Upload your PDF

Drag and drop file here  
Limit 200MB per file • PDF

Browse files

Or enter text manually:

Enter URLs to extract text from (comma-separated):

Select your education level:  
High School

Select the severity of the questions:  
Easy

Number of questions to generate:  
5

Generate Questions

Generate Summary

**Need assistance with a topic?**

Enter a topic or question you'd like help with:

Figure 2: Application Interface

## B Appendix B: Sample prompt

```
prompts > E tough_questions.txt
1 Generate a list of {num_questions} tough questions for {level} students related to the given context.
2 These questions should involve a complex understanding of the topic, requiring critical analysis, comparison, or synthesis of multiple
3 concepts.
4 The answers to these questions should be detailed, nuanced, and demand a high level of explanation, often integrating multiple perspectives
5 or addressing the topic's broader implications.
6 Generate questions only, no explanations, or answers or additional text is required.
```

Figure 3: Sample prompt of hard question

## C Appendix C: Sample Questions Generated

Input used to generate the below questions: source URL

| Level→/<br>Education↓ | High School                                                                               | Bachelors                                                                                                                                                                                                                                                | Masters                                                                                                                                                                                                                         | PhD                                                                                                                                                                                                                                                                                |
|-----------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Easy</b>           | Which company released GPT-4, currently the most powerful LLM?                            | What is fine-tuning in the context of LLMs?                                                                                                                                                                                                              | What is the primary application of Zero-shot LLMs?                                                                                                                                                                              | What does LLaMa stand for in the context of LLMs?                                                                                                                                                                                                                                  |
| <b>Medium</b>         | What are some common applications or industries where LLMs have shown significant impact? | What is the primary difference between a pre-trained LLM and a fine-tuned LLM, and how does it impact their performance on specific tasks?                                                                                                               | Explain the relationship between LLMs and the phenomenon of “hallucinations”. What are the causes and consequences of hallucinations in LLMs, and how can they be mitigated or addressed?                                       | Explain how the growth of open-source LLMs like LLaMa 2 is influencing the development of proprietary models like GPT-4, and what implications this has for the broader AI research community.                                                                                     |
| <b>Difficult</b>      | Evaluate the environmental impact of large-scale LLM training and deployment.             | Discuss the ethical implications of developing and deploying Large Language Models that generate content that could be harmful or misleading (e.g., fake news, propaganda), and propose strategies for mitigating their potential negative consequences. | How do LLMs like BERT, PaLM 2, and LLaMa 2 differ in their architecture and design choices that impact their performance on natural language processing (NLP) tasks, and what are the implications for real-world applications? | Design an experiment to investigate the effects of adversarial attacks on the performance of a state-of-the-art LLM, considering metrics such as perplexity, accuracy, and interpretability, and discuss potential implications for LLM-based applications in high-stakes domains. |

Table 3: Example questions on LLMs across education levels and difficulty tiers

## D Appendix D: Sample Summary

Input used to generate the below summary: source URL

### Document Summary:

Large Language Models (LLMs) are revolutionizing various fields by enabling advanced language understanding and generation. This document explores LLM concepts, applications, challenges, and future implications.

#### Core Concepts:

- **Transformers:** The foundation of LLMs lies in transformers, a deep learning architecture that excels at processing sequential data like text. Transformer models leverage attention mechanisms to analyze relationships between words or phrases within a sentence. This allows them to comprehend context more effectively than traditional recurrent neural networks (RNNs).
- **Training Data:** LLM training requires massive datasets of text and code. The more data, the better the model learns language patterns, but also introduces biases if the data reflects societal prejudices.
- **Language Understanding & Generation:** LLMs excel at tasks like:
  - **Text Summarization:** Condensing lengthy texts into concise summaries.
  - **Question Answering:** Accurately answering questions posed based on provided context.
  - **Translation:** Translating text between languages with high accuracy.
  - **Code Generation:** Generating functional code in various programming languages.

**Applications Across Industries:** The diverse range of LLMs' capabilities fuels widespread applications:

- **Customer Service:** Automated chatbots and virtual assistants can provide quick and efficient customer service.
- **Education & Research:** LLMs help students learn better with personalized tutoring, and researchers analyze vast datasets for scientific breakthroughs.
- **Healthcare:** LLMs assist in patient diagnosis based on medical records, drug development, and understanding complex medical literature.

**Challenges & Considerations:** While powerful, LLMs are not without their challenges:

- **Ethical Concerns:** Bias, privacy breaches, and misuse of LLMs for malicious purposes require careful ethical considerations.
- **Resource Intensive:** Training and deploying LLMs require significant computational resources (e.g., vast data storage, high-performance GPUs).

**Future Implications:** LLMs promise a future with:

- **Democratized Access to AI:** As LLM development progresses, their use will become more accessible, empowering individuals and organizations.
- **Enhanced Creativity & Productivity:** LLMs can assist in content creation, research, and problem-solving, boosting productivity across industries.

**Conclusion:** LLMs represent a paradigm shift in AI. Their ability to understand and generate human language opens doors for innovation and societal advancement. Addressing ethical concerns and ensuring responsible development will be crucial to realizing the full potential of LLMs and shaping their future impact on society.

Figure 4: Sample summary generated by our application